

BAYER IN INDIA: INTELLECTUAL PROPERTY EXPROPRIATION?¹

Peter M. Bican and Quynh Nhu Truong wrote this case solely to provide material for class discussion. The authors do not intend to illustrate either effective or ineffective handling of a managerial situation. The authors may have disguised certain names and other identifying information to protect confidentiality.

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In 2012, the Intellectual Property Appellate Board (IPAB) decided to grant Indian drug manufacturer Natco Pharma Limited (Natco) a compulsory licence (CL). This licence allowed Natco to legally manufacture, sell and distribute a copy of Bayer AG's (Bayer) blockbuster cancer drug,² Nexavar, in India at a fraction of the original price. This was bad news not only for Bayer but also for the entire pharmaceutical industry. Natco's victory in securing the CL marked a paradigm shift that would change the way research-and-development (R&D)-intensive multinational corporations conducted business.

Bayer Group's chief executive officer (CEO), Marijn Dekkers, declared, "We have a lot of trouble with a ruling like this, which at its core states that India does not value patent rights at all."³ It meant that pharmaceutical innovation could not be effectively protected by conventional intellectual property rights in emerging markets. Hence, the core of the pharmaceutical industry's business model was called into question: If ideas and inventions could not be protected by intellectual property rights, was there an incentive for firms to innovate at all? Would the CL granted to Natco trigger similar rulings in India or elsewhere? Would intellectual expropriation spread through the pharmaceutical industry like cancer?

THE GLOBAL PHARMACEUTICAL INDUSTRY

The global pharmaceutical industry could be divided into two major groups of players: generic manufacturers and research-based pharmaceutical producers, with the first group largely imitating and reproducing the products developed by the second group.

The research-based pharmaceutical industry generated US\$956 billion⁴ in 2011, with total sales projected to reach \$1.2 trillion by 2016.⁵ At that time, the market was dominated by Pfizer Inc. (7.1 per cent market share), Novartis AG (5.9 per cent market share) and Merck & Co. Inc. (4.9 per cent market share). Bayer's healthcare division secured about 2 per cent of the world's pharmaceutical market.⁶ These major players in the research-based pharmaceutical industry were sometimes collectively referred to as "Big Pharma."⁷

The United States remained the biggest market for pharmaceutical producers in 2010. Japan ranked second but represented only one-third of the U.S. volume. At the same time, mature markets had shrunk in sales, while emerging economies in Africa, Asia and Latin America had picked up speed.⁸

Decision makers in the research-based pharmaceutical industry had to face significant new challenges. Overall annual industry growth had been 6.1 per cent from 2007 to 2011, but forecasted growth from 2012 to 2016 stood at only 4.5 per cent.⁹ The ongoing aftermath of the 2008 global economic crisis, increased competition from generic manufacturers, additional pressure from local governments to slash prices, and greater regulatory scrutiny by patent approval agencies had all negatively affected large pharmaceutical firms. In addition, their heavy dependency on patent protection held the biggest threat for Big Pharma. The impending patent expiry of 13 out of 20 key blockbuster drugs — including Lipitor, Viagra and Nexium — was about to further damage their profits.¹⁰

Decoding, healing and eradicating major diseases such as cancer and HIV/AIDS had long been the holy grail of the research-based pharmaceutical industry. Cancer was one of the leading causes of death, accounting for 7.6 million deaths worldwide in 2008.¹¹ According to one study, India alone accounted for approximately 556,400 of the total deaths caused by cancer in 2010.¹² The most common forms of cancer included lung, stomach, liver, colorectal, breast and cervical cancers. With Nexavar, Bayer was able to ease patients' suffering from kidney and liver cancer.

Most research-focused pharmaceutical companies were either located in the United States or conducted the majority of their R&D activity there. Typically, the development of a drug in developed markets took about 10 to 15 years. In 2012, the cost of developing a single drug was over \$1.3 billion,¹³ up from only \$138 million in 1975. The delay in developing and marketing a drug was not only due to internal and product-related factors; in fact, a major share of cost and time could be attributed to the regulatory approval process, which, as a local right, had to be undergone for each country where approval was sought.

Drugs could be developed, produced and approved in various dosage forms, routes of administration and strengths; these forms, routes and strengths might vary for the same drug based on the medical condition being treated or customer preference and convenience. For instance, patients might prefer oral administration versus injections or suppositories, capsules over pills, taking one dose per day versus two, etc.

In the United States, the U.S. Food and Drug Administration (FDA) was the government body that administered the drug approval process and granted marketing approval. The process of development and FDA approval could be summed up in six steps: early phase research; pre-clinical testing and Investigational New Drug Application (IND); clinical trials; regulatory review; scale-up to manufacturing; and post-marketing surveillance (see Exhibit 1). The patents themselves were granted by the U.S. Patent and Trademark Office (USPTO).

In India, the Indian Patent Office (IPO) administered the Indian Patent Act, including granting patents. Marketing authorization (i.e., the regulatory approval to sell the drug in India, comparable to FDA approval in the U.S.) was granted by India's central regulatory body, the Central Drugs Standard Control Organization (CDSCO). The drugs controller general (India) was the head of the CDSCO. There was no patent linkage in India (as opposed to the U.S.), which meant that marketing approvals could be formally applied for via the Drugs and Cosmetics Act 1940 and Drugs and Cosmetics Rules 1945. Scientific data, safety and efficacy were sufficient grounds to grant a marketing authorization, regardless of the patent

status. Once the CDSCO granted regulatory approval, firms had to apply for manufacturing permission at the state level before they could begin production.¹⁴

THE PHARMACEUTICAL INDUSTRY IN INDIA

Until 2005, only manufacturing processes could be patented in India, leaving products largely unprotected. Thus, drugs could be copied as long as their production involved the application of a different process. This led Indian pharmaceutical firms to specialize in reverse-engineering innovator drugs instead of conducting their own R&D in-house. These so-called “generic” drugs were unlicensed, low-cost copies that were launched once the innovator drug had lost its patent protection.¹⁵ In 2008, Indian pharmaceutical producers were not only manufacturing low-cost medicines for the local market but also supplying more than 20 per cent of the total generic drug market worldwide.¹⁶ For domestic drug manufacturers, the transition from a reverse-engineering focus to a more R&D-driven business model was not expected to occur in the near future.

In 2012, the Indian population spent \$15.6 billion¹⁷ on pharmaceutical products, equivalent to \$11 per capita annually (see Exhibit 2). Total health spending in India amounted to 4.1 per cent of the gross domestic product (GDP) in 2010, against an Organization for Economic Cooperation and Development (OECD) average of 9.5 per cent. India’s healthcare infrastructure was very weak: per 1,000 persons there were only 0.7 physicians, one nurse and 0.9 hospital beds available, compared to 2.4 physicians, 11 nurses and 3.1 hospital beds in the United States.¹⁸ Additionally, health insurance was rare and 80 per cent of all healthcare expenses in India were financed out of pocket.¹⁹ As affluence and life expectancy improved with economic growth, the therapeutic needs of the Indian nation changed as well. An increasing number of people were diagnosed with illnesses often occurring in developed countries, such as diabetes and cardiovascular diseases. It was predicted that these diverging therapy areas and higher buying power would drive future pharmaceutical sales in India.

However, the market was expected to remain highly price-sensitive since 81.1 per cent of the population in India lived on less than \$2.50 a day and as much as 32.7 per cent subsisted on less than \$1.25 a day.²⁰ In stark contrast to those living at the bottom of the pyramid, around 100 million people in India were, as Ian Read, CEO of Pfizer, remarked, “[as wealthy or wealthier than] the average European or American.”²¹ This wealth disparity was blatantly visible in cities like Mumbai, which had numerous shanty towns as well as fancy shopping malls selling foreign luxury brands at prices even higher than those of Paris, Rome or Munich.

INTELLECTUAL PROPERTY

Intellectual property (IP) played an especially important role in the pharmaceutical industry. As a local right, IP encompassed a wide range of creations, involving “inventions, literary and artistic works, symbols, names, images, and designs used in commerce,”²² and could be protected by various IP rights like patents, trademarks, industrial designs, copyrights (associated with the creative arts) and trade secrets.

Trademarks were a commonly used form of IP protection. Trademarks existed to protect consumers from the likelihood of confusion and came into force through use in commerce (registration was not mandatory but generally advised). Trademark protection extended from word marks to designs, sounds or symbols; even colours and shapes could be trademarked. Theoretically, when used in commerce, trademarks could endure indefinitely (see Exhibit 3).

In the pharmaceutical industry, patents were regarded as the most powerful IP tool to protect pharmaceutical products.²³ A patent was defined by the World Intellectual Property Organization (WIPO) as “a document, issued, upon application, by a government office, which describes an invention and creates a legal situation in which the patented invention can normally only be exploited (manufactured, used, sold, imported) with the authorization of the owner of the patent.”²⁴

Patent rights were so-called “negative” rights — that is, the patentee owned the right to exclude others from using, making or selling the patented invention for a limited period of time, usually 20 years. In addition, patents were territorial rights and therefore only valid in the country in which they were granted to the inventor (see Exhibit 3).

Patents and India

The first Indian patent legislation was derived from British patent law and dated back to the Act VI of 1856.²⁵ In 1911, the Indian Patents and Designs Act replaced all previous regulations on patents. This legislation did not change until India achieved independence in 1947, when the entire patent system was completely overhauled to match the needs of the newly founded country. It was over 20 years before the committee constituted by the government to review the law finally agreed on the contents of IP protection laws and passed the Patents Act of 1970, which replaced the 1911 Act. This law protected certain manufacturing processes for drugs, pharmaceuticals and foods, which greatly benefited the rise of the generic drugs industry in subsequent years. However, with India joining the World Trade Organization (WTO) in 2005,²⁶ and its accession to the WTO Trade-related Aspects of Intellectual Property Rights (TRIPS) Agreement, India also had to uniformly guarantee patent protection for 20 years. Accordingly, a new product patent regime came into effect on January 1, 2005. India had previously joined the WIPO in 1975 and had signed the Patent Cooperation Treaty (PCT), an international agreement that focused on patent searches and filing patent applications, in 1998.

According to the TRIPS Agreement,²⁷ an invention in India qualified for a patent if it was novel, industrially applicable and showed an “inventive step” (i.e., an advancement of either technological or economic significance that must be “not obvious” to experts of that trade). Moreover, Indian patent laws demanded the “local working” of a patent. If it failed to work in India, a CL might be granted. The beneficiary of a CL had “the right to perform acts covered by the exclusive right under an authorization given by a government authority against the will of the owner of the patent.”²⁸ CLs generally fell into two categories and were only granted in the event of the abuse of a patent or in the public interest — that is, in relation to public health, national defence or economic development.²⁹ Anyone in India could apply for a CL three years after a patent approval if the patent owner had not complied with the requirements set by the Indian Patent Act. CLs were legal under Article 31 of the TRIPS Agreement, which permitted all member countries to grant authorization to third parties to exploit the invention in question. In 2001, the Doha Ministerial Declaration on TRIPS and Public Health, a pending amendment to the TRIPS Agreement, explicitly supported CLs for pharmaceuticals as long as generics were produced for the local market and a “reasonable royalty” was paid to compensate the original inventor.³⁰ Hence, the TRIPS Agreement provided India with broad flexibility to individually set up and interpret requirements to justify CLs.³¹

FROM THE INVENTOR OF ASPIRIN

Bayer AG, founded in 1863, was one of the oldest German research-based holding companies active in the pharmaceutical and chemical industry. In 2012, Bayer’s global operations were carried out by a total

of 110,500 employees in over 350 different locations and total group sales amounted to \$51.7 billion (see Exhibits 4 and 5). At the end of 2012, Bayer offered more than 20,000 products worldwide and owned roughly 76,000 patents and patent applications, ranking among the top 50 corporate patent applicants worldwide.³²

The company had three main business areas: Bayer HealthCare, Bayer CropScience and Bayer MaterialScience. It was a well-known inventor and innovator, most famous for the invention of Aspirin. Bayer HealthCare was a purely research-based company. All products produced and sold by Bayer HealthCare were so-called innovator drugs, which the inventor was exclusively entitled to sell under a brand name. Generic manufacturers were only entitled to sell the drug under the name of the compound (its generic name). For example, the generic (bio-equivalent) version of Aspirin (the branded drug, whose name was trademarked by Bayer) could be sold under its generic name, acetylsalicylic acid.

Bayer had a long history in India. The company first established its presence in 1896. Bayer India was targeting \$1.3 billion in sales through organic and inorganic growth by 2015.³³ Bayer HealthCare (BHC) India operated in three business areas: pharmaceuticals, medical care and animal health.³⁴ The pharmaceuticals segment consisted of two business units focusing on prescription products: General Medicine, primarily comprising women's healthcare products and cardiovascular health products; and Specialty Medicine, comprising medicines that were mainly prescribed by specialist physicians.³⁵ The products in the latter segment, such as Glucobay (an anti-diabetic drug), Yasmin/Yaz (a market leader in the oral contraceptive market), Baycip, Bayer's Tonic and Canestan (an anti-fungal product) held key positions in the therapeutic areas that they served. Nexavar (for kidney and liver cancer) was part of the Specialty Medicine unit.³⁶

THE BLOCKBUSTER DRUG

The anti-cancer drug Nexavar was based on the active ingredient sorafenib tosylate.³⁷ The first patent was granted by the USPTO on June 26, 2007, for the treatment of primary kidney cancer (U.S. Registration Number 7,235,576). Two additional patents were awarded to the sorafenib family on April 1, 2008 (U.S. 7,351,834), and on March 1, 2011 (U.S. 7,897,623). Bayer owned two trademarks for Nexavar, one being the name "Nexavar" as well as its logo depicting blue and ochre sickles (U.S. 3,537,095).³⁸

Nexavar was not a cure for cancer; however, it slowed the growth and spread of cancer cells. According to several clinical trials, Nexavar could prolong a kidney cancer patient's life by four to five years and the life of a liver cancer patient by up to eight months. The red, round 200-milligram tablets could only be obtained with a prescription. It was to be taken orally at a dosage of two tablets, twice a day, without food.

Competition in the market among drugs treating advanced kidney cancer was fierce. The Kidney Cancer Association listed five other drugs in the angiogenesis inhibitor category along with Nexavar, including Pfizer's Sutent, which was approved by the FDA just one month after Nexavar. Since then, four additional drugs had been approved in the same therapeutic area: Torisel (Wyeth), Afinitor (Novartis), Votrient (GlaxoSmithKline) and Inlyta (Pfizer). For the most common type of liver cancer that Nexavar treated, it was the only non-surgical treatment available at that time.³⁹

Regulatory approval for the treatment of advanced renal cell carcinoma (RCC — primary kidney cancer) was granted by the FDA in December 2005 (FDA N021923). Additionally, Nexavar was awarded orphan drug designation (i.e., categorization as a medicine used in rare diseases) by the FDA in April 2006. In

2007, Nexavar received regulatory approval for the treatment of hepatocellular carcinoma (HCC — primary liver cancer), making it the first approved medication to treat HCC. By April 2013, Nexavar was approved in more than 80 countries for the treatment of liver cancer and in more than 90 countries for the treatment of advanced kidney cancer (see Exhibit 6).⁴⁰ It was one of Bayer's blockbuster drugs in 2012, with \$1.04 billion in global sales (see Exhibits 7 and 8).

Nexavar in India

Nexavar entered the Indian patent registration process in 2001, using an international application under WIPO's Patent Cooperation Treaty (PCT). It was granted regulatory approval in India by the CDSCO on July 31, 2007. The drug was not patented with the IPO until March 3, 2008 (Patent Number 215758).⁴¹ Nexavar was sold at a price of \$5,500 (INR300,000) for a month's supply of 120 tablets, equivalent to the price Bayer charged its American patients at that time.⁴²

Natco: Generic Manufacturer under CL

Natco, a reputed generic drug manufacturer, was one of India's fastest-growing firms in the pharmaceutical industry.⁴³ The company had initially filed for a voluntary licence (VL) for Nexavar from Bayer but was quickly turned down.⁴⁴ On July 29, 2011, Natco applied for a CL under Section 84(1) of the Indian Patent Act and Rule 96 of the Patent Rules (2003) before the controller general of patents. It proposed to sell its generic version of Nexavar at a price of \$160 (INR8,800) for a month's supply — a discount of over 97 per cent on Bayer's price. On March 9, 2012, the controller general granted the first-ever CL in India to Natco. He decided that the low availability of Nexavar, its high price and a lack of "working" of the patent in India were violations of Section 84(1)(a), (b) and (c) of the Indian Patent Act, respectively. As a result, Natco was granted the CL, committing the company to pay 7 per cent of the drug's royalties⁴⁵ to Bayer. The ruling represented a stunning defeat for the IP rights community and a historical victory for the Indian pharmaceutical industry (see Exhibit 9).

THE CONTROLLER GENERAL'S RULING

Reasonable Requirements of the Public

The controller general justified his ruling by asserting that Bayer underestimated the demand for the drug and did not make the drug readily available to the public. The Indian authorities agreed with Natco that Bayer's calculated number of incidents was too conservative, thus leading to an insufficient amount of Nexavar imported to India. Second, Nexavar was supposedly only available in pharmacies attached to premium hospitals in metropolitan areas (such as Mumbai, Chennai, Kolkata and Delhi) and was often out of stock even in those areas. Therefore, the controller general concluded that Bayer had failed to fulfil Section 84(1)(a) of the Indian Patent Act.

Reasonably Affordable Price

Section 84(1)(b) allowed the inventor to charge a reasonable market price for a patented drug. Bayer argued that drug prices must mirror R&D efforts and costs associated with the invention of the drug.

In the case of Nexavar, however, the term “reasonableness” was calculated only in regard to Bayer and not in the context of the needs of the general public. A month’s supply of Nexavar cost \$5,500, which was 41 times the per capita GDP in India.⁴⁶ Consequently, the controller general ruled that the drug was unreasonably priced.

Working in Indian Territory

The controller general concluded that since Bayer did not produce Nexavar in India, mere importing did not fulfil the local working requirement of Section 84(1)(c) of the Patent Act. A combined reading of Section 83(c) and (f), which stated that patent holders were obliged to transfer and spread the technology nationally and internationally, further confirmed the controller general’s interpretation. This technology transfer could have been realized by either local manufacturing or licensing. Even though Bayer manufactured other products in India, the firm did not produce Nexavar in India and had refused to out-license to local firms such as Natco. Thus, the controller general was convinced that Bayer did not work the invention properly in India.

THE IMPACT OF THE CONTROLLER GENERAL’S RULING

Upon approval of the CL, Natco was legally allowed to manufacture and market its own version of sorafenib without Bayer’s authorization. Natco agreed to pay Bayer a quarterly royalty of 7 per cent on its net sales. Additional conditions restricted the production of the drug solely to Natco facilities (i.e., no outsourcing was permitted) and its sales to India only; it could not be exported to other countries. Product branding and packaging were required to be completely different from Nexavar.⁴⁷ In response, Bayer filed for a petition to block Natco’s generic version, with one of its arguments against the ruling being that Bayer did indeed provide significant quantities of the drug to eligible patients. In 2008, Bayer had launched a patient assistance program in cooperation with Indian doctors to give patients better access to Nexavar by providing it at 10 per cent of its market price. According to Bayer, more than 73 per cent of patients deemed eligible for the cancer drug benefited from this program in 2012.⁴⁸ The firm argued that it had supplied the cancer drug to patients in need, even in smaller towns, but a lack of “trained healthcare staff and infrastructure, cultural acceptability of treatment, accessibility of healthcare facilities, and quality of care all [played] a role,”⁴⁹ which the controller general did not consider in his decision.

STAKEHOLDER PERSPECTIVES ON THE CONTROLLER GENERAL’S RULING

The Indian Institute of Patent and Trademark advised the Indian government to strongly reconsider the “trade-offs between making vital medicines available at a reasonable cost and important IP issues,”⁵⁰ as India had been able to attract a considerable amount of foreign direct investment⁵¹ since it began to protect IP in 2005. The body argued that the Nexavar ruling could make foreign firms reluctant to conduct business and launch new drugs in India.⁵² Ranjit Shahani, CEO of Novartis India, reacted to the court order as follows: “We are convinced that innovations need to be protected by patents if we should continue our efforts in medical advancements.”⁵³

Most of the large global news agencies and newspapers, including Reuters, The Wall Street Journal, The Economist and TIME, noted that the ruling had negative repercussions for the pharmaceutical industry worldwide. A column in a prominent German business publication argued that the IPAB ruling was rather unrewarding.⁵⁴ The author criticized the order as being a sign of “gross lobbying” in favour of the Indian generics pharmaceuticals industry. “Roles are clearly divided in this case,” the journalist wrote, “On the

one hand, we have the ‘evil pharma industry’ that does not mind charging the poor astronomical drug prices. On the other hand, we have the Indian courts acting as the saviour of the poor.” The author posited that a poor population would not be saved with affordable medicine alone, which reiterated an argument made by Bayer on the importance of improving medical infrastructure by training medical staff and building hospitals that were able to cater to the medical needs of the people. Power plays with patents seemed unlikely to lead to sustainable improvements for the medical situation in emerging economies.

In contrast, the World Health Organization reacted positively to the ruling: “India has taken a good political stand on compulsory licenses and we respect that move [... that intends to] keep public health interests above any other interests.”⁵⁵ Leena Menghaney of Médecins Sans Frontières, an international medical humanitarian aid organization, also welcomed the decision of the IPAB. She envisioned that the ruling would generate more CLs so that cheaper medicines could be offered in other developing countries as well.⁵⁶

Yet other industry experts believed the controller general’s decision could have unintended consequences. Stephen Oxley, head of pharmaceuticals at KPMG in the United Kingdom, expressed his concerns:

No one would argue with the benefits of having medicines cheaply and more readily available to poorer parts of the world population, but we still have to bear in mind the realities of drug production. If the appetite of the major pharmaceutical players to invest in R&D (for) new products aimed at diseases endemic to the third world is diminished by announcements such as these, then the pipelines of new drugs in those disease classes will dry up.⁵⁷

Indian generic drug manufacturers expected large business opportunities to emerge from CLs. With the majority of the Indian pharmaceutical industry specializing in reverse-engineering and low-cost manufacturing, these firms were now encouraged to seek out high-priced, high-value patented drugs and apply for additional CLs.⁵⁸ CLs for drugs made by Western pharmaceutical firms Bristol-Myers Squibb, Roche and Novartis were already lined up and were likely to be granted by Indian authorities.⁵⁹

BIG PHARMA IN INDIA — THE FUTURE?

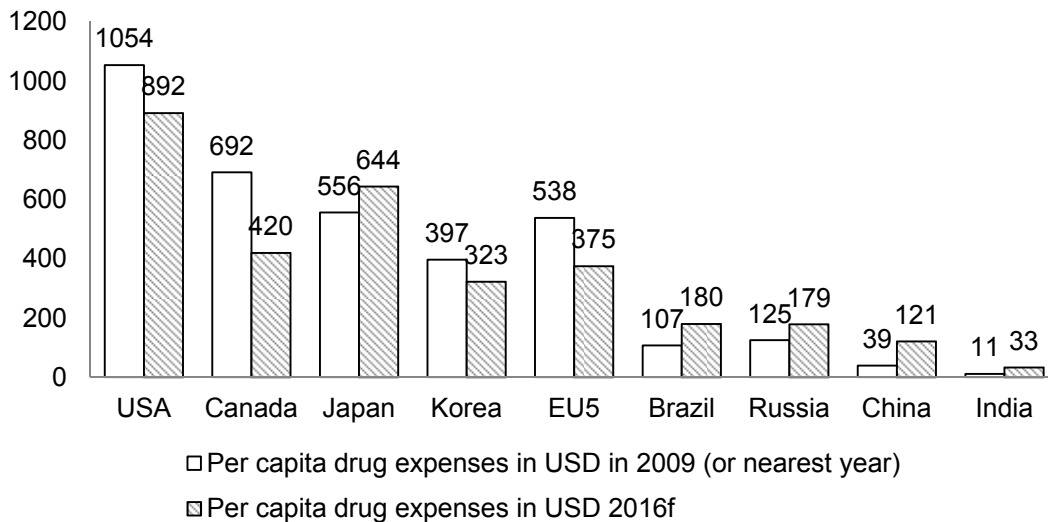
The ruling and its repercussions appeared to be growing out of control and threatening to spread across the global pharmaceutical industry like cancer. It was a wake-up call for Bayer as well as the entire industry, forcing key decision makers to reassess their business situations. Would they have to adapt their existing strategic IP management in emerging markets to the fast-changing environment? Which factors would determine Bayer’s successful resurrection from this incident? Would Bayer have to withdraw from markets with weak IP protection if long-term profitability could not be guaranteed? How would Bayer’s Western competitors respond to the ruling in India? What consequences did it have for Bayer’s Indian competitors, who were primarily generic drug manufacturers? What would the long-term implications be for patients in India?

The authors are from WHU – Otto Beisheim School of Management, Germany. Peter M. Bican would like to acknowledge the encouraging discussions with his PhD advisors Holger Ernst of the WHU-Otto Beisheim School of Management and James G. Conley of the Kellogg School of Management (Northwestern University) that shaped his interest and understanding in IP and innovation management.

EXHIBIT 1: DRUG DEVELOPMENT AND APPROVAL

Early Phase Research	Before starting any research scientists must thoroughly understand the disease they are aiming to treat. Often, commercial R&D teams collaborate with academia, start-up firms and independent research labs to exploit a larger knowledge base. Researchers often identify a single “target” molecule and then begin to create and screen 5,000 to 10,000 potential compounds in search for the one “lead compound” that later becomes the marketed drug. Initial tests are conducted with promising drugs to evaluate the benefits of the compound to existing therapies. Manufacturing costs are also estimated at this stage. If a promising compound is found, a patent may be filed at the USPTO. This process takes from four to six years.
Pre-clinical Testing & Investigational New Drug Application (IND)	This stage takes from one to three years and involves intensive lab and animal testing to affirm that the substances are efficacious and safe for human testing. Before clinical trials on humans can begin, an IND Application with all available information (results of pre-clinical tests, side effects, manufacturing, etc.) must be filed with and approved by the FDA.
Clinical Trials	This trial phase involving humans takes from two to 10 years in total. The number of human volunteers increases with the progress made in the trial phase. In Phase I, the safety, dosage and activity pattern of the drug are tested on a small number of healthy volunteers. In Phase II, the optimal dose strength and possible side effects are tested. These controlled trials are conducted with a larger group of volunteers, with half taking the drug and half taking a placebo. Phase III involves a large sample of volunteers (about 1,000 to 5,000) to confirm safety, efficacy and additional long-term side effects.
Regulatory Review	A New Drug Application (NDA), including all available information and trial results, must now be submitted to the FDA in order to gain approval to commercialize the drug (i.e., marketing authorization). The authorities can approve or reject the application, or request more information. This stage takes between six months and two years. This linkage of marketing approval with prevalent patent status is referred to as patent linkage. Approved drugs are listed in the so-called “Orange Book,” and no other person is granted a marketing approval (i.e., permission to legally produce, distribute and sell) if a drug is still registered in the Orange Book.
Scale-up to Manufacturing	Out of 5,000 to 10,000 compounds initially tested, only one, on average, successfully transitions through the R&D pipeline and can be produced on a large scale. Sometimes new production facilities are set up and must meet strict FDA guidelines.
Post-marketing Surveillance	Even after the product launch, companies are obliged to continuously monitor and provide information to the FDA on how the drug is received by patients. This phase also involves a Phase IV trial and studies involving subgroups of patients (e.g., the elderly or children).

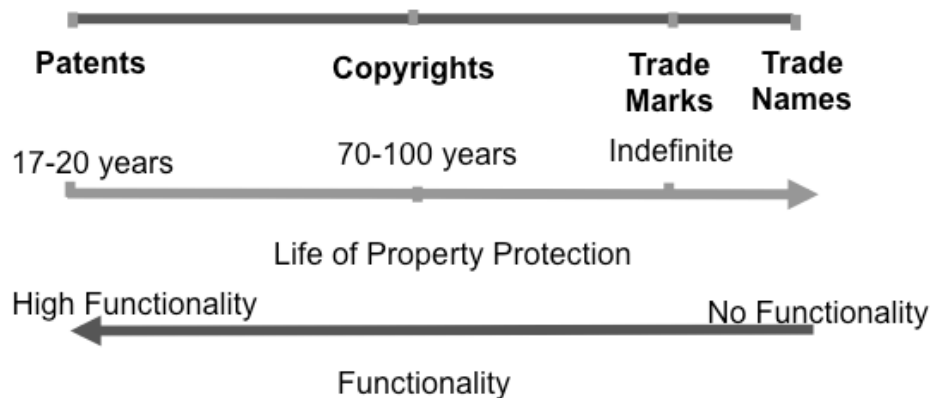
Source: Case authors, based on “The Pharmaceutical Industry and Global Health — Facts & Figures 2012,” *International Federation of Pharmaceutical Manufacturers & Associations (IFPMA)*, December 2012, www.ifpma.org/fileadmin/content/Publication/2013/IFPMA_-_Facts_And_Figures_2012_LowResSinglePage.pdf, accessed June 14, 2013.

EXHIBIT 2: PER CAPITA SPENDING ON MEDICATION (2009 AND 2016 FORECAST)

Note: Annual per capita spending in US\$; EU5 describes the United Kingdom, Germany, France, Italy and Spain.

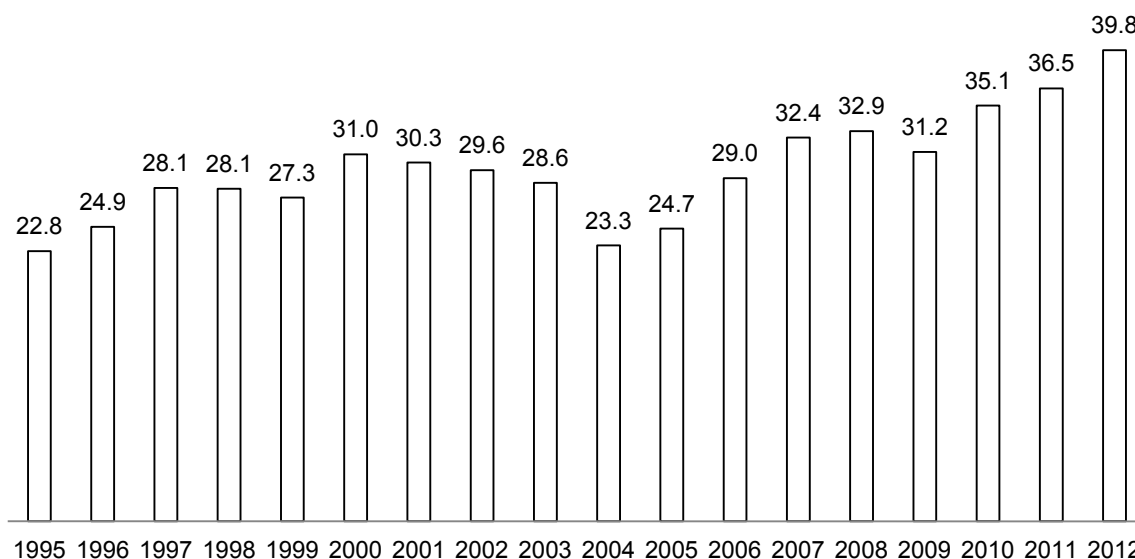
Source: "Lebensstandard und Kaufkraft für Arzneimittel ausgewählter Länder 2010," *Statista*, 2013, www.de.statista.com/statistik/daten/studie/224031/umfrage/lebensstandard-und-kaufkraft-fuer-arzneimittel-ausgewaehlter-laender/, accessed May 4, 2013.

"The Global Use of Medicines: Outlook through 2016," *IMS Institute for Healthcare Informatics*, July 2012, www.imshealth.com/deployedfiles/ims/Global/Content/Insights/IMS%20Institute%20for%20Healthcare%20Informatics/Globa%20Use%20of%20Meds%202011/Medicines_Outlook_Through_2016_Report.pdf, accessed May 29, 2013, p. 6.

EXHIBIT 3: IP CONTINUUM OF PROTECTION

Source: Case authors, adapted from James. G. Conley, Peter. M. Bican and Ernst Holger, "Value Articulation — A Framework for the Strategic Management of Intellectual Property," 2012, presented at the EPO Workshop "Intellectual Property Management: In Search of New Practices, Strategies and Business Models," June 4-5, 2012, Munich, Germany.

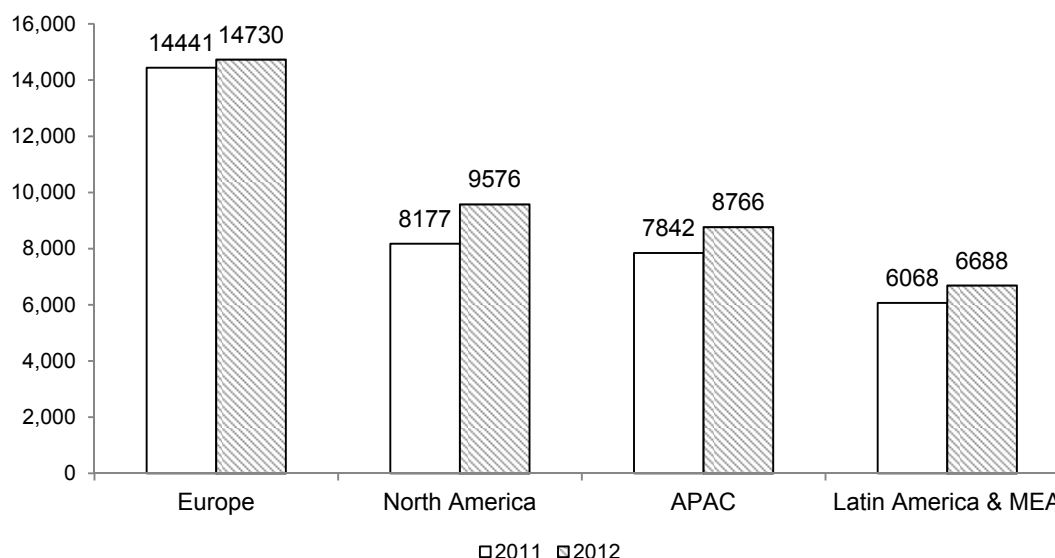
EXHIBIT 4: BAYER GROUP SALES DEVELOPMENT (1995-2012)



Note: Sales in billions of euro (€).

Source: "Revenue of Bayer AG from 1995 to 2012 (in million euros)," *Statista*, 2012, www.statista.com/statistics/263778/revenue-of-bayer-ag-since-1995/, accessed February 19, 2013.

EXHIBIT 5: BAYER GROUP'S DISTRIBUTION OF SALES BY REGION (2011-2012)

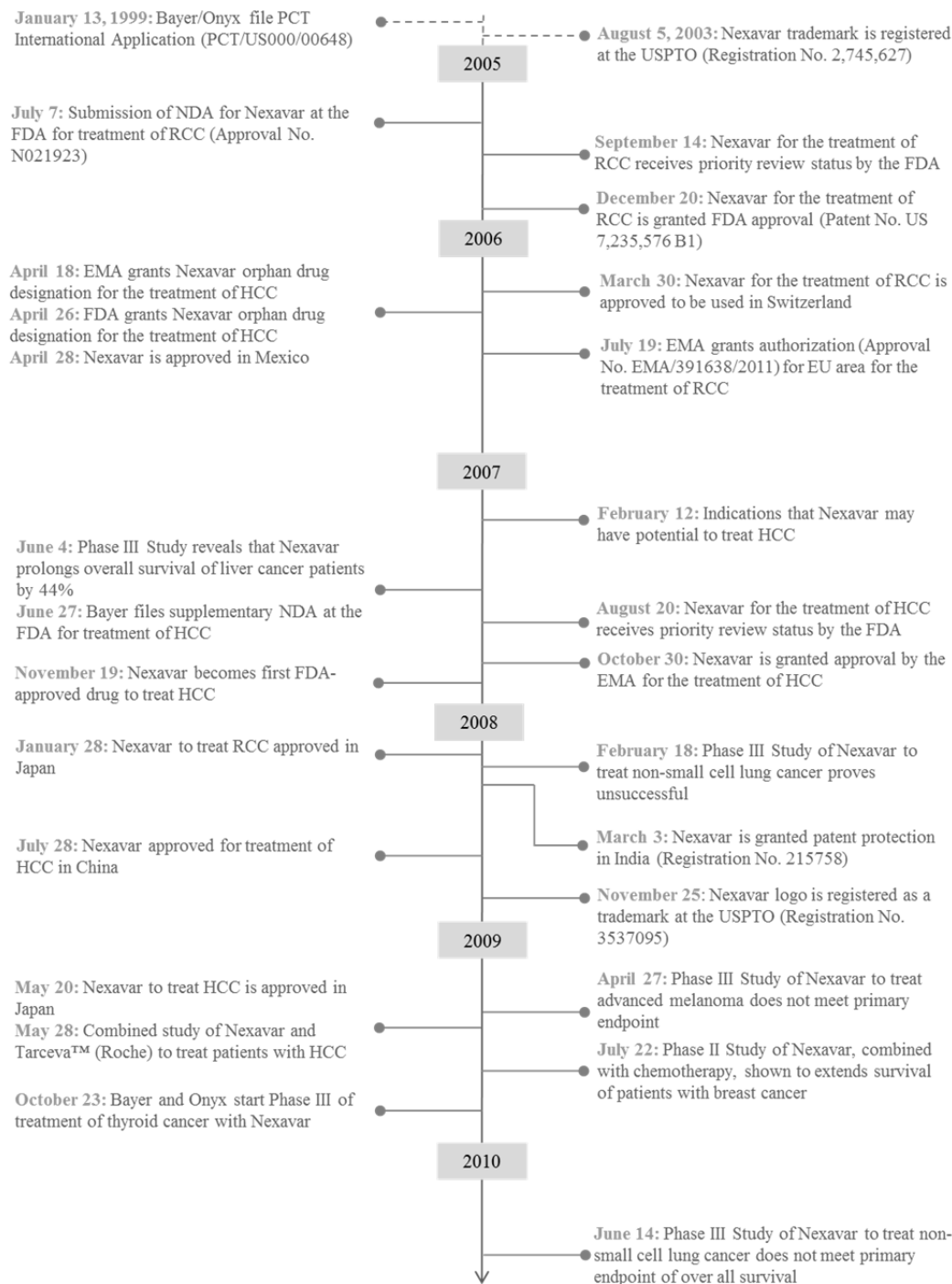


Europe saw growth of 2%; North America – 17.2%, APAC – 11.8%, and Latin America & MEA – 10.2%.

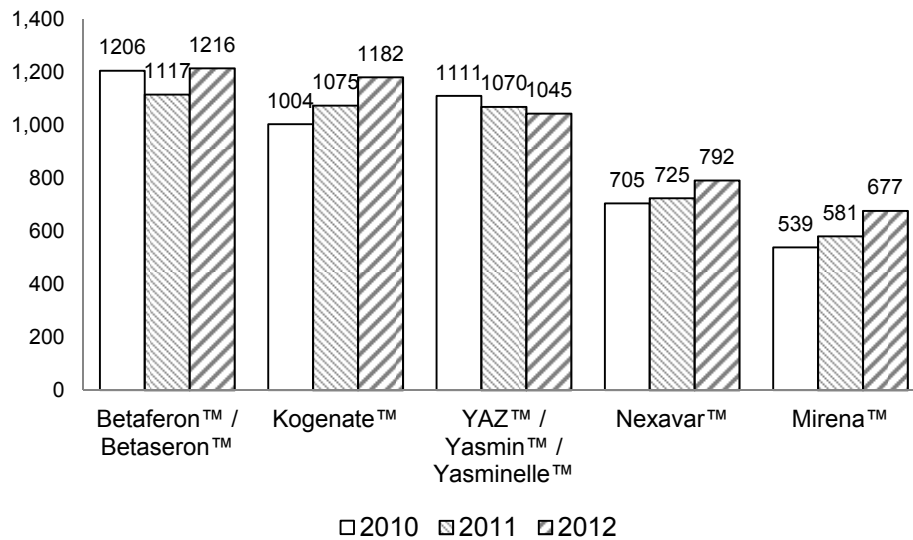
Note: Growth rates are calculated from 2011 to 2012; sales in billions of euro (€).

Source: "Umsatz der Bayer AG nach Region in den Jahren 2011 und 2012 (in Millionen Euro)," *Statista*, 2012, www.de.statista.com/statistik/daten/studie/247090/umfrage/umsatz-der-bayer-ag-nach-region/, accessed March 1, 2013.

EXHIBIT 6: NEXAVAR TIMELINE

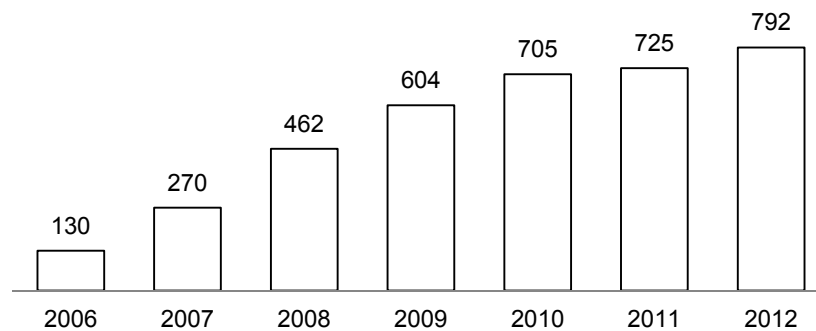


Sources: Created by authors based on Bayer AG Annual Reports 2005-2012, www.bayer.com/en/annual-reports.aspx, accessed February 12, 2013; www.drugs.com/search.php?searchterm=nexavar, accessed: February 18, 2013; www.ema.europa.eu/ema/index.jsp?curl=pages/medicines/human/medicines/000690/human_med_000929.jsp&mid=WC0b01ac058001d124, accessed: February 22, 2013; www.investor.bayer.de/en/nc/news/archive, accessed April 9, 2013; http://pharma.bayer.com/scripts/pages/en/news_room/news_room/archive/newsroom_archive40.php, accessed: April 9, 2013.

EXHIBIT 7: BAYER GROUP'S TOP 5 PHARMA PRODUCTS SOLD (2010-2012)

Note: Numbers represent 2012 sales figures in millions of euro (€).

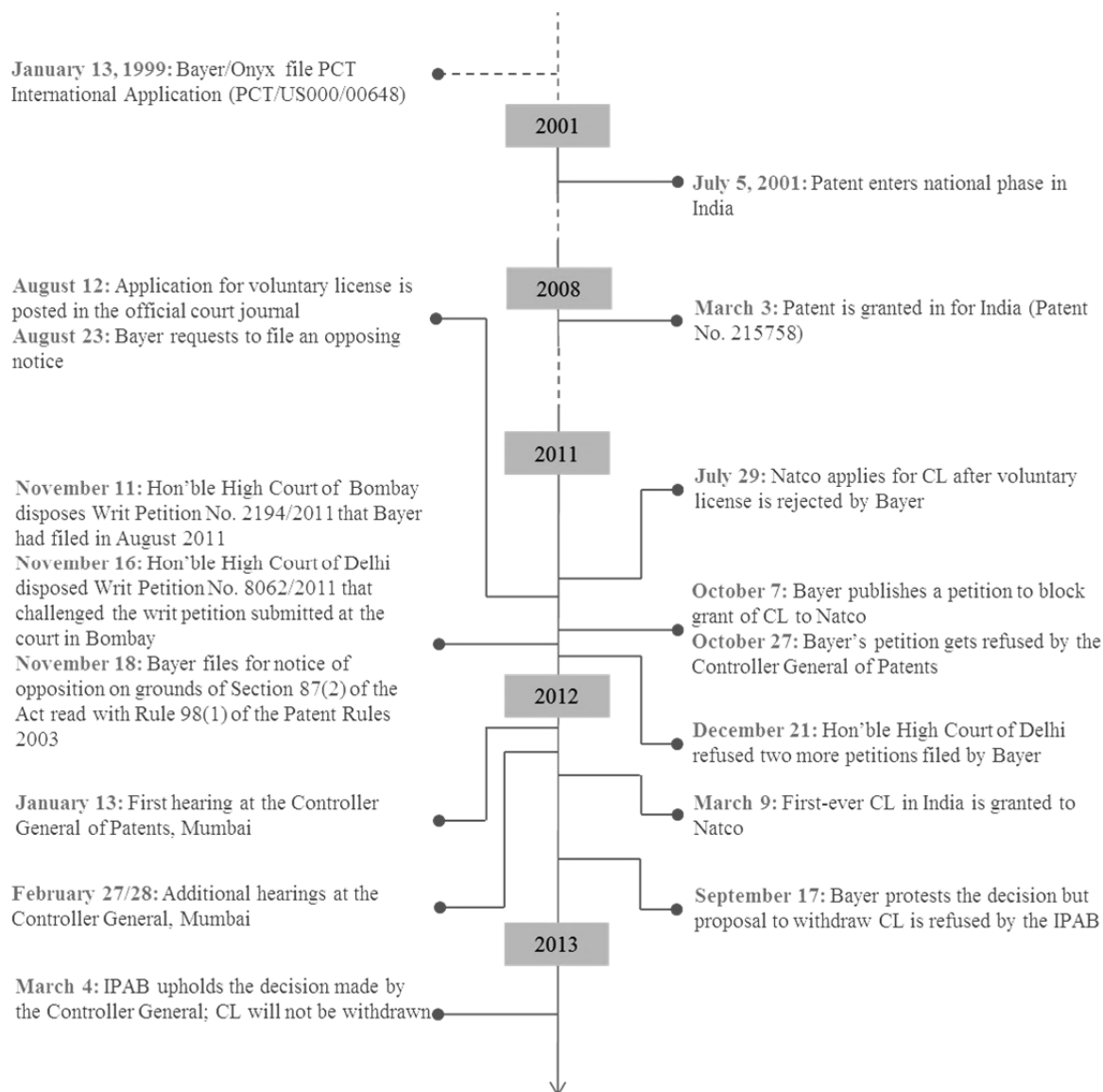
Sources: "Bayer AG's Top Pharmaceutical Products from 2010 to 2012, Based on Revenue (in Million Euros)," Statista.com, www.statista.com/statistics/263787/revenues-of-bayers-top-pharmaceutical-products/, accessed June 30, 2013.

EXHIBIT 8: NEXAVAR GLOBAL SALES DEVELOPMENT (2006-2012)

Note: Nexavar sales figures in millions of euro (€).

Sources: Case authors, based on data from Bayer Annual Reports 2006-2012, [Bayer AG](http://www.bayer.com/en/Annual-Reports.aspx), www.bayer.com/en/Annual-Reports.aspx, accessed June 30, 2013.

EXHIBIT 9: NEXAVAR INDIA TIMELINE



Source: Case authors.

ENDNOTES

- ¹ This case has been written on the basis of published sources only. Consequently, the interpretation and perspectives presented in this case are not necessarily those of Bayer Group or any of its employees.
- ² Blockbuster drugs are usually defined as drugs with yearly revenue in excess of US\$1 billion, see with: James G. Conley, Peter M. Bican and Neil Wilkof, "Study on Patents and the Public Domain (II)," *WIPO CDIP Discussion Paper*, 2013, www.wipo.int/edocs/mdocs/mdocs/en/cdip_12/cdip_12_inf_2_rev.pdf, accessed November 30, 2013.
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